

FUNDATA, Romania

WMO: 153010

Lat: 45.4314N

Lon: 25.2717E

Elev: 1386

StdP: 85.75

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.7	(c) -13.7	(d) -18.9	(e) 0.8	(f) -12.1	(g) -16.5	(h) 1.0	(i) -11.1	(j) 14.2	(k) -7.6	(l) 11.9	(m) -7.4	(n) 4.0	(o) 230	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.8	(c) 23.3	(d) 17.2	(e) 22.0	(f) 16.1	(g) 20.7	(h) 15.5	(i) 18.3	(j) 22.2	(k) 17.2	(l) 20.6	(m) 16.2	(n) 19.5	(o) 2.6	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.9	(b) 14.3	(c) 20.7	(d) 15.8	(e) 13.3	(f) 19.3	(g) 14.9	(h) 12.5	(i) 18.2	(j) 58.0	(k) 22.2	(l) 54.1	(m) 20.6	(n) 51.0	(o) 19.7	(p) 21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.8	(b) 7.7	(c) 6.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.8	(b) 7.7	(c) 6.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-4.9	-2.2	-0.1	4.5	9.9	14.1	15.0	15.7	11.2	6.3	2.1	-2.4		
	DBStd	8.39	5.33	4.99	4.79	5.51	3.80	3.25	3.02	3.36	4.71	5.10	5.08	4.71		
	HDD10.0	2159	463	342	314	177	47	6	2	2	44	138	239	385		
	HDD18.3	4593	722	575	572	414	262	131	108	92	216	372	487	643		
	CDD10.0	630	0	0	1	13	43	128	157	180	81	24	2	0		
	CDD18.3	23	0	0	0	0	0	3	6	12	3	0	0	0		
	CDH23.3	40	0	0	0	0	0	3	11	18	8	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	0	1	1	0	0	0		
(14) Wind	WSAvg	2.9	3.4	3.4	3.2	3.0	2.7	2.5	2.4	2.3	2.5	2.8	3.4	3.3		
(15) Precipitation	PrecAvg	885	46	42	52	70	105	119	113	96	74	63	53	52		
	PrecMax	1240	125	83	93	144	195	217	197	207	147	136	107	122		
	PrecMin	575	8	7	13	35	35	52	12	31	20	4	4	8		
	PrecStd	166	27	19	22	28	41	45	48	46	35	32	27	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	10.5	13.5	18.3	20.8	23.7	24.6	25.5	23.9	19.6	15.1	9.4		
		MCWB	2.8	5.6	6.6	10.7	13.5	16.2	18.3	18.3	18.3	11.2	8.7	3.0		
	2%	DB	5.3	8.1	10.9	16.3	18.9	21.7	22.7	23.4	20.9	17.3	12.4	7.3		
		MCWB	2.2	3.5	4.9	9.7	12.7	15.7	16.8	17.4	14.6	10.5	7.4	2.8		
	5%	DB	3.8	6.1	8.5	14.3	17.2	20.4	21.2	22.1	19.1	15.3	10.4	5.5		
		MCWB	1.6	2.3	3.8	8.6	12.0	15.2	16.2	16.4	13.9	9.9	6.5	2.7		
	10%	DB	2.2	4.4	6.5	12.2	15.5	19.1	19.8	20.7	17.4	13.3	8.9	3.7		
		MCWB	0.2	1.6	2.9	7.4	11.3	14.6	15.5	15.8	12.8	8.6	6.0	1.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	6.9	7.1	11.4	15.0	17.7	19.2	19.8	18.5	13.1	9.5	5.6		
		MCDB	6.2	9.6	12.1	16.7	18.2	21.8	23.2	22.8	23.0	16.1	13.0	6.7		
	2%	WB	3.0	4.6	5.4	10.3	13.6	16.6	18.1	18.5	16.1	11.3	8.2	3.7		
		MCDB	4.6	7.1	9.7	15.5	17.3	20.1	21.5	22.3	19.3	15.6	11.4	5.8		
	5%	WB	2.0	3.1	4.2	9.2	12.5	15.7	16.9	17.5	14.4	10.3	7.1	2.9		
		MCDB	3.5	5.5	8.1	13.6	16.4	19.1	19.7	21.3	17.8	14.7	9.9	5.0		
	10%	WB	0.4	2.0	3.1	8.0	11.5	15.0	16.0	16.2	13.3	9.2	6.3	1.7		
		MCDB	1.9	3.7	6.1	11.7	15.0	18.6	19.0	19.8	16.7	12.7	8.7	3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	5.8	6.2	7.1	6.8	7.1	6.9	6.8	6.7	6.3	5.5	5.2		
		MCDBR	6.9	7.6	8.6	9.0	8.3	8.0	8.0	7.7	8.0	8.2	6.9	6.6		
	5% WB	MCWBR	4.6	4.9	5.0	5.0	5.1	4.9	4.9	4.9	4.5	4.8	4.1	4.4		
		MCWBR	6.9	6.7	7.9	8.2	8.2	7.6	7.4	7.4	7.1	7.7	6.1	5.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.268	0.293	0.329	0.379	0.363	0.364	0.376	0.388	0.351	0.320	0.299	0.271		
		taud	2.381	2.326	2.318	2.244	2.340	2.357	2.323	2.287	2.379	2.466	2.479	2.430		
	Edn at Noon	Edn at Noon	851	889	900	879	904	901	885	859	866	850	809	810		
		Edh at Noon	73	94	110	130	122	120	123	123	103	81	67	63		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.40	2.22	3.29	4.29	5.23	5.66	5.76	5.21	3.82	2.64	1.54	1.15		
	RadStd	RadStd	0.15	0.28	0.34	0.47	0.51	0.51	0.49	0.43	0.37	0.34	0.16	0.14		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	+0.59	N/A	N/A	N/A	+0.73	N/A	N/A	-128	-174	+76	+24

Nomenclature: See separate page